

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.12_k-1)

Metric

$$ds^2 = -x_3^2 \sinh(x_1)^2 dx_2 \otimes dx_2 + x_3^2 dx_1 \otimes dx_1 + \left(\frac{9x_3^4}{\Lambda^2 x_3^8 + 12\Lambda m x_3^5 + 6\Lambda x_3^6 + 3(2\Lambda e^2 + 3)x_3^4 + 36mx_3^3 + 18(2m^2 + e^2)x_3^2 + 36mx_3e^2 + 9e^4} \right) dx_3 \\ \otimes dx_3 + \left(\frac{\Lambda^2 x_3^8 + 12\Lambda m x_3^5 + 6\Lambda x_3^6 + 3(2\Lambda e^2 + 3)x_3^4 + 36mx_3^3 + 18(2m^2 + e^2)x_3^2 + 36mx_3e^2 + 9e^4}{9x_3^4} \right) dx_4 \otimes dx_4$$

Coordinates

$$(x^\mu) = (x_2, x_1, x_3, x_4)$$

Metric Functions

$$\Sigma = \sinh(x_1)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} x_3 \sinh(x_1) \right) dx_2 + \left(-\frac{1}{2} \sqrt{2} x_3 \right) dx_1$$

$$n = \left(\frac{1}{2} \sqrt{2} x_3 \sinh(x_1) \right) dx_2 + \left(\frac{1}{2} \sqrt{2} x_3 \right) dx_1$$

$$m = \left(-\frac{3\sqrt{2}}{2\left(\Lambda x_3^2 + \frac{6m}{x_3} + \frac{3e^2}{x_3^2} + 3\right)} \right) dx_3 + \left(-\frac{1}{6}i\sqrt{2}\Lambda x_3^2 - \frac{1}{2}i\sqrt{2} - \frac{i\sqrt{2}m}{x_3} - \frac{i\sqrt{2}e^2}{2x_3^2} \right) dx_4$$

$$\bar{m} = \left(-\frac{3\sqrt{2}}{2\left(\Lambda x_3^2 + \frac{6m}{x_3} + \frac{3e^2}{x_3^2} + 3\right)} \right) dx_3 + \left(\frac{1}{6}i\sqrt{2}\Lambda x_3^2 + \frac{1}{2}i\sqrt{2} + \frac{i\sqrt{2}m}{x_3} + \frac{i\sqrt{2}e^2}{2x_3^2} \right) dx_4$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2x_3 \sinh(x_1)} \right) dx_2 + \left(-\frac{\sqrt{2}}{2x_3} \right) dx_1$$

$$n = \left(-\frac{\sqrt{2}}{2x_3 \sinh(x_1)} \right) dx_2 + \left(\frac{\sqrt{2}}{2x_3} \right) dx_1$$

$$m = \left(-\frac{1}{6}\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\sqrt{2} - \frac{\sqrt{2}m}{x_3} - \frac{\sqrt{2}e^2}{2x_3^2} \right) dx_3 + \left(-\frac{3ix_3^2}{\sqrt{2}\Lambda x_3^4 + 6\sqrt{2}mx_3 + 3\sqrt{2}x_3^2 + 3\sqrt{2}e^2} \right) dx_4$$

$$\bar{m} = \left(-\frac{1}{6}\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\sqrt{2} - \frac{\sqrt{2}m}{x_3} - \frac{\sqrt{2}e^2}{2x_3^2} \right) dx_3 + \left(\frac{3ix_3^2}{\sqrt{2}\Lambda x_3^4 + 6\sqrt{2}mx_3 + 3\sqrt{2}x_3^2 + 3\sqrt{2}e^2} \right) dx_4$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2x_3 \sinh(x_1)} \partial_{x_2} - \frac{\sqrt{2}}{2x_3} \partial_{x_1}$$

$$\Delta = -\frac{\sqrt{2}}{2x_3 \sinh(x_1)} \partial_{x_2} + \frac{\sqrt{2}}{2x_3} \partial_{x_1}$$

$$\delta = -\frac{1}{6}\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\sqrt{2} - \frac{\sqrt{2}m}{x_3} - \frac{\sqrt{2}e^2}{2x_3^2} \partial_{x_3} - \frac{3ix_3^2}{\sqrt{2}\Lambda x_3^4 + 6\sqrt{2}mx_3 + 3\sqrt{2}x_3^2 + 3\sqrt{2}e^2} \partial_{x_4}$$

$$\bar{\delta} = -\frac{1}{6}\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\sqrt{2} - \frac{\sqrt{2}m}{x_3} - \frac{\sqrt{2}e^2}{2x_3^2} \partial_{x_3} + \frac{3ix_3^2}{\sqrt{2}\Lambda x_3^4 + 6\sqrt{2}mx_3 + 3\sqrt{2}x_3^2 + 3\sqrt{2}e^2} \partial_{x_4}$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{1}{6} \sqrt{2} \Lambda x_3 - \frac{\sqrt{2} m}{2 x_3^2} - \frac{\sqrt{2} e^2}{2 x_3^3}$$

$$\beta = -\frac{1}{6} \sqrt{2} \Lambda x_3 + \frac{\sqrt{2} m}{2 x_3^2} + \frac{\sqrt{2} e^2}{2 x_3^3}$$

$$\tau = \frac{1}{6} \sqrt{2} \Lambda x_3 + \frac{\sqrt{2} m}{x_3^2} + \frac{\sqrt{2}}{2 x_3} + \frac{\sqrt{2} e^2}{2 x_3^3}$$

$$\pi = -\frac{1}{6} \sqrt{2} \Lambda x_3 - \frac{\sqrt{2} m}{x_3^2} - \frac{\sqrt{2}}{2 x_3} - \frac{\sqrt{2} e^2}{2 x_3^3}$$

$$\epsilon = -\frac{\sqrt{2} \cosh (x_1)}{4 x_3 \sinh (x_1)}$$

$$\gamma = -\frac{\sqrt{2} \cosh(x_1)}{4 x_3 \sinh(x_1)}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{5}{36} \Lambda^2 x_3^2 + \frac{\Lambda m}{3 x_3} + \frac{\Lambda e^2}{6 x_3^2} - \frac{2 m^2}{x_3^4} + \frac{1}{2 x_3^2} - \frac{5 m e^2}{x_3^5} - \frac{e^2}{x_3^4} - \frac{9 e^4}{4 x_3^6}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{5}{36} \Lambda^2 x_3^2 - \frac{1}{3} \Lambda - \frac{\Lambda m}{3 x_3} - \frac{\Lambda e^2}{18 x_3^2} - \frac{1}{6 x_3^2} - \frac{m e^2}{3 x_3^5} - \frac{e^4}{4 x_3^6}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{1}{18} \Lambda^2 x_3^2 + \frac{\Lambda e^2}{9 x_3^2} + \frac{4 m^2}{x_3^4} + \frac{2 m}{x_3^3} + \frac{1}{3 x_3^2} + \frac{20 m e^2}{3 x_3^5} + \frac{2 e^2}{x_3^4} + \frac{5 e^4}{2 x_3^6}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"